

Bølger og bølgefunksjonar

- Ulike former for bølger
- Bølgelikninga i ein dimensjon
- Harmoniske bølger
- Ståande bølger
- Bølgepakker
- Schrödingerlikninga
- Bølgefunksjonen
- Heisenbergs uskarpleiksrelasjon
- Dobbeltspalteeksperimentet

Bølgelikninga

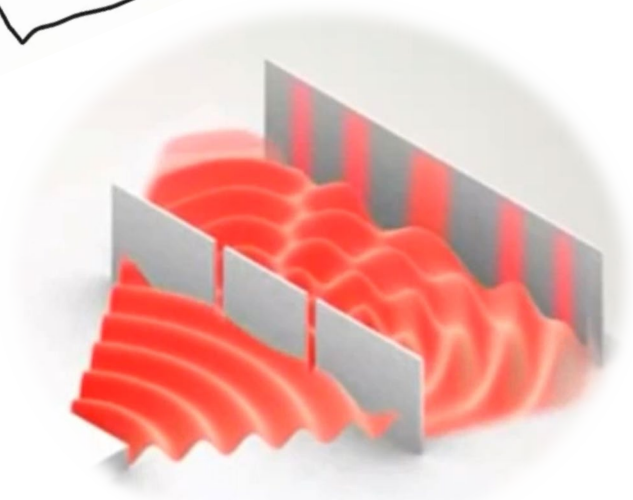
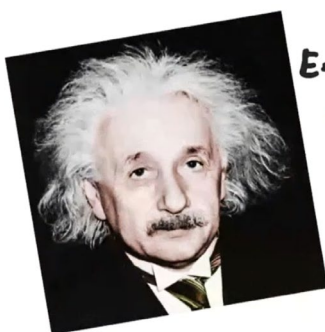
$$\frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} = \frac{\partial^2 y}{\partial x^2}$$

$$\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + U(x) \right] \psi(x) = E \psi(x)$$

$$\hat{H} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + U(x) \right]$$

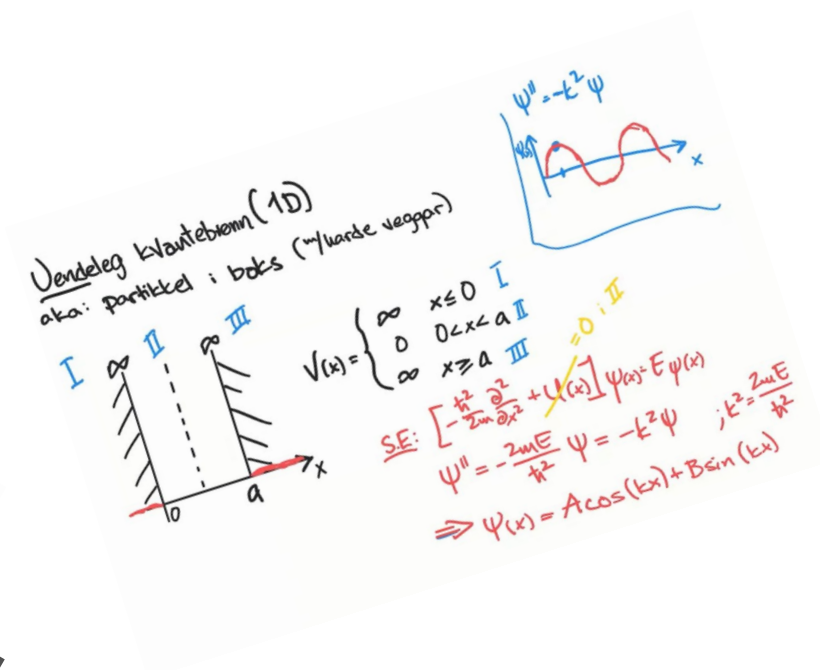


$$u(x,t) = A \cos(kx - \omega t + \phi)$$

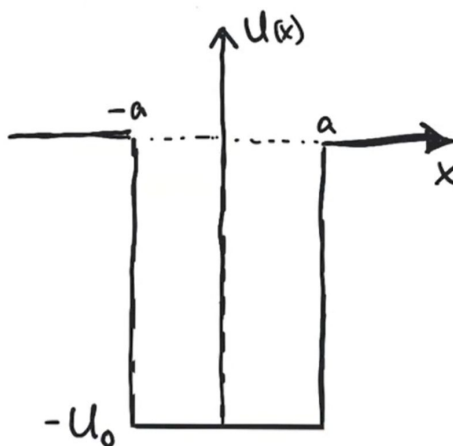


Kvantebrønner

- Bundne og spreidende tilstandar
- Uendeleg djup kvantebrønn i 1D
- Målepostulatet
- Endeleg kvantebrønn
- Tunnelling
- Perturbasjonsteori

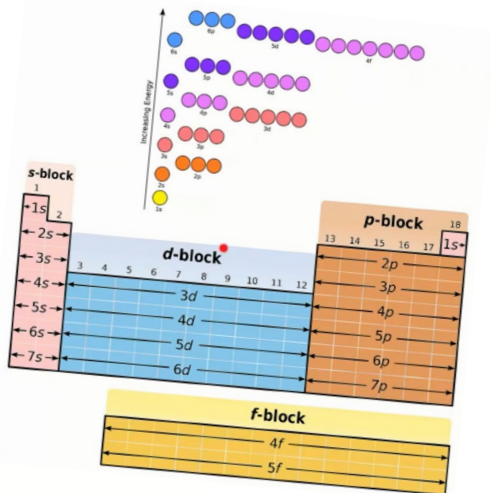


$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

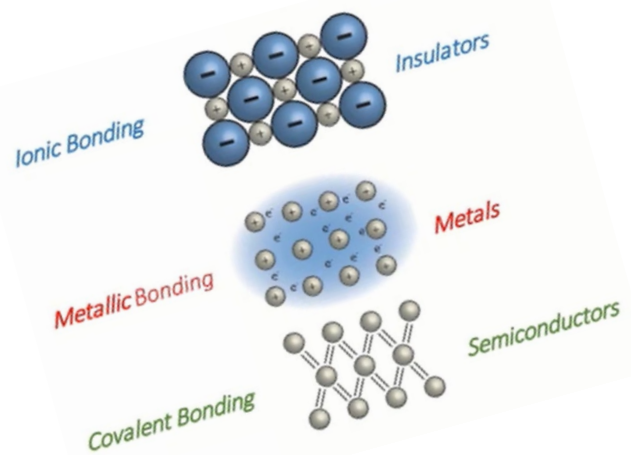
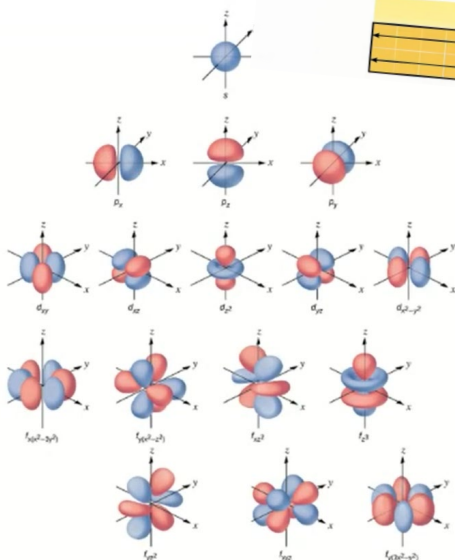


Atom og bindingar

- Kvantebrønn i 3D (partikkel i boks)
- Degenerasjon
- Hydrogenatomet
- Det periodiske system
- Elektronkonfigurasjon
- Bindingar i faste stoff
- Dobbel kvantebrønn
- Det interatomære potensiale

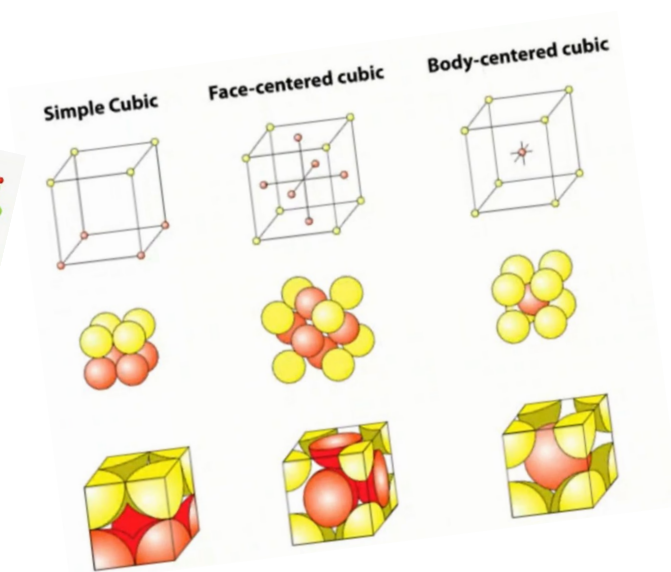
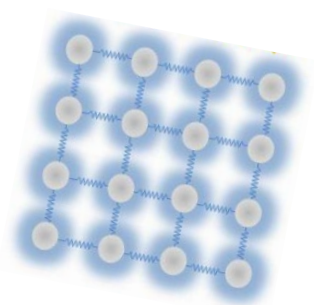
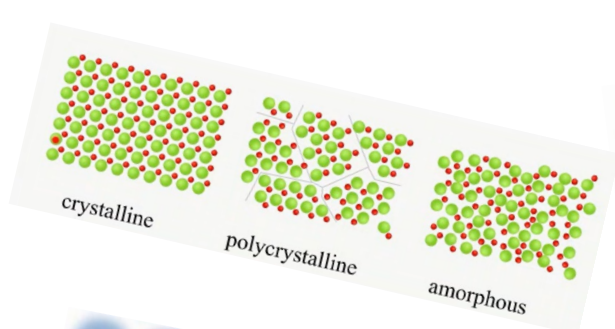
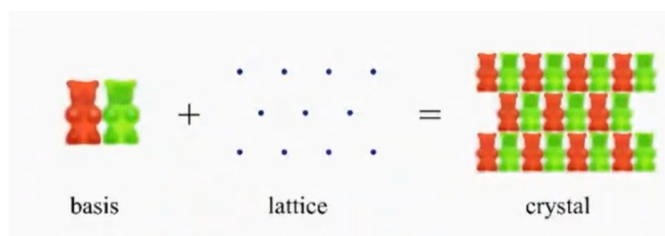


1A	2A	3A	4A	5A	6A	7A	8A
1 H 1s ¹							2 He 1s ²
3 Li 2s ¹	4 Be 2s ²	5 B 2s ² 2p ¹	6 C 2s ² 2p ²	7 N 2s ² 2p ³	8 O 2s ² 2p ⁴	9 F 2s ² 2p ⁵	10 Ne 2s ² 2p ⁶
11 Na 3s ¹	12 Mg 3s ²	13 Al 3s ² 3p ¹	14 Si 3s ² 3p ²	15 P 3s ² 3p ³	16 S 3s ² 3p ⁴	17 Cl 3s ² 3p ⁵	18 Ar 3s ² 3p ⁶



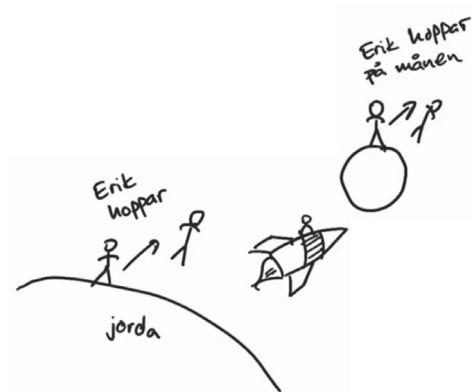
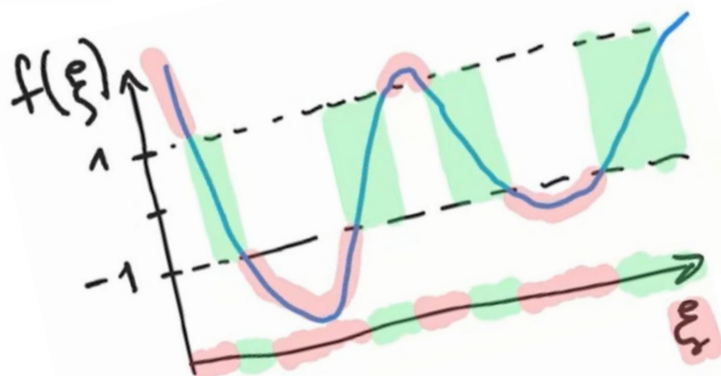
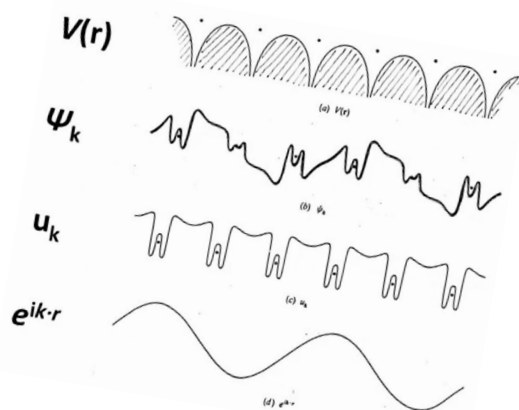
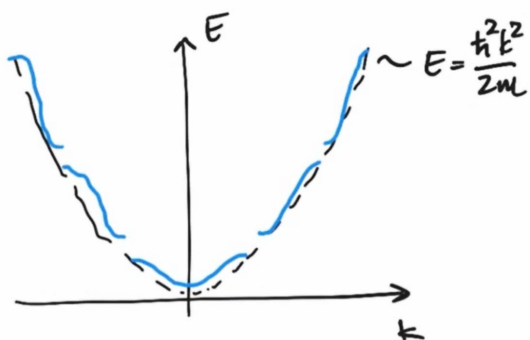
Krystallografi og fonon

- Faste stoff
- Krystallar, basis og gitter
- Einingsceller og Bravaisgitter
- Kubiske gitter (fcc, bcc, sc)
- Miller indeksar
- Wigner-Seitz cella
- Reelt og resiprokt gitter
- Fonon
- Debyemodellen
- Varmeleiing, varmekapasitet og termisk ekspansjon



Kronig –Penney modellen

- Kronig-Penney modellen – som modell for 1D gitter
- Forenklinger i Kronig-Penney modellen
- Bloch's teorem
- Løysingane til Kronig-Penney modellen
- Dispersjonsrelasjonen, $E(k)$
- Effektiv masse



Båndstruktur

- Båndstruktur i 1. Brillouinsone
- Fermi-Dirac fordelinga
- Tilstandstettleik (DOS)
- Elektron og hull i båndstrukturen
- Båndstruktur i 3D
- Direkte og indirekte båndgap
- Forenkla bånddiagram

